

Virtual.c

```
GNU nano 8.7.1
#include <stdio.h>
int main() {
int x = 10;

printf("Value of x = %d\n", x);
printf("Virtual address of x = %p\n", (void *)&x);
return 0;
}
```

```
Child exited with status 0
userhasini@DESKTOP-QDSCC4:~$ nano virtual.c
userhasini@DESKTOP-QDSCC4:~$ gcc virtual.c -o virtual
userhasini@DESKTOP-QDSCC4:~$ ./virtual
Value of x = 10
```

Dynamicmemory:

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>

int main() {

    int *p;

    p = malloc(5 * sizeof(int));

    if (p == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }

    for (int i = 0; i < 5; i++) {
        p[i] = (i + 1) * 10;
    }

    for (int i = 0; i < 5; i++) {
        printf("%d ", p[i]);
    }

    printf("\n");

    free(p);

    return 0;
}
```

```
Virtual address of x = 0x7ffffce744f4
userhasini@DESKTOP-QDSCC4:~$ nano dynamicmemory.c
userhasini@DESKTOP-QDSCC4:~$ gcc dynamicmemory.c -o dynamicmemory
userhasini@DESKTOP-QDSCC4:~$ ./dynamicmemory
10 20 30 40 50
userhasini@DESKTOP-QDSCC4:~$ |
```

Address transition:

```

GNU nano 8.7.1
#include <stdio.h>

int main() {

    int page_table[] = {3, 5, 1, 7};
    int page_size = 100;

    int virtual_address = 250;

    int page = virtual_address / page_size;
    int offset = virtual_address % page_size;

    int frame = page_table[page];

    int physical_address = frame * page_size + offset;

    printf("Virtual Address = %d\n", virtual_address);
    printf("Page Number    = %d\n", page);
    printf("Offset         = %d\n", offset);
    printf("Frame Number    = %d\n", frame);
    printf("Physical Address = %d\n", physical_address);

    return 0;
}

```

```

userhasini@DESKTOP-QDSCC4:~$ nano virtual.c
userhasini@DESKTOP-QDSCC4:~$ nano dynamicmemory.c
userhasini@DESKTOP-QDSCC4:~$ nano adresstransition.c
userhasini@DESKTOP-QDSCC4:~$ gcc adresstransition.c -o adresstransition
userhasini@DESKTOP-QDSCC4:~$ ./adresstransition
Virtual Address = 250
Page Number    = 2
Offset         = 50
Frame Number    = 1
Physical Address = 150
userhasini@DESKTOP-QDSCC4:~$ |

```

Address:

```

GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>

int global = 10;
int bss_variable;

int main() {

    int local = 20;

    int *heap = malloc(sizeof(int));
    *heap = 30;

    printf("Code address : %p\n", (void *)main);
    printf("Data address : %p\n", (void *)&global);
    printf("BSS address : %p\n", (void *)&bss_variable);
    printf("Heap address : %p\n", (void *)heap);
    printf("Stack address : %p\n", (void *)&local);

    free(heap);

    return 0;
}

```

```

userhasini@DESKTOP-QDSCC4:~$ nano address.c
userhasini@DESKTOP-QDSCC4:~$ gcc address.c -o address
userhasini@DESKTOP-QDSCC4:~$ ./address
Code address : 0x636fb92851a9
Data address : 0x636fb9288010
BSS address : 0x636fb9288018
Heap address : 0x636fba4ae010
Stack address : 0x7ffdc0f9915c
userhasini@DESKTOP-QDSCC4:~$

```